

# Fundamentals of Electric Circuits

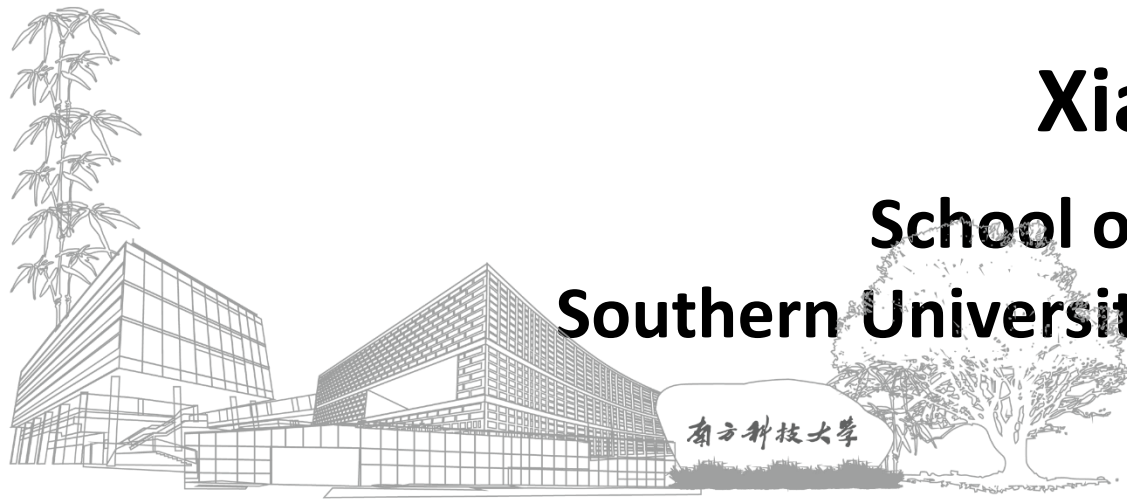
## 电路基础

### Lecture7b: First-Order Circuits

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# Outline

- Introduction
- The Source-Free *RC* Circuit
- The Source-Free *RL* Circuit
- Singularity Functions
- Step Response of an *RC* Circuit
- Step Response of an *RL* Circuit

# Introduction

- We will examine two simple but critical circuits: **RC circuit** and **RL circuit**. Applying Kirchhoff's laws to analyze *RC* and *RL* circuits, we get **differential equations (微分方程)**, rather than the algebraic ones for purely resistive circuits
  - Since the differential equations are of first order, *RC* and *RL* circuits are called **first-order circuits (一阶电路)**
  - Capacitors and inductors have voltage-current (or current-voltage) relationships described by **differentials**, i.e.,  $i_C = C \cdot dv_C/dt$ ,  $v_L = L \cdot di_L/dt$
- The circuits can be excited by **initial conditions (初始状态)** of the **storage elements** and **independent sources**. We will examine both the **source-free (零输入)** and **switched source (外加激励)** scenarios
  - 1) Source-Free *RC* Circuit; 2) Source-Free *RL* Circuit
  - 3) Step response of an *RC* circuit; 4) Step response of an *RL* circuit

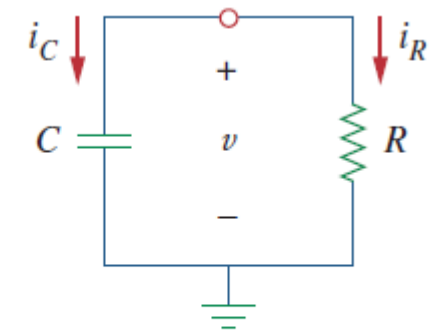
# The Source-Free $RC$ Circuit

- A **source-free** (零输入)  $RC$  circuit occurs when its dc source is suddenly disconnected. The energy stored in the capacitor is released to the resistors
- Consider a series combination of a resistor and an initially charged capacitor
  - Note that the resistor and capacitor may be the equivalent resistor and equivalent capacitor of complex resistive and capacitive networks
- Assume that the initial voltage of the charged capacitor is  $v(0) = V_0$ . Applying KCL at the node on the top, we have

$$i_C + i_R = 0 \rightarrow C \frac{dv}{dt} + \frac{v}{R} = 0 \rightarrow \frac{dv}{dt} + \frac{v}{RC} = 0$$

- The equation is a first-order differential equations and can be solved by the separable equation method, the solution is

$$\frac{dv}{v} = -\frac{dt}{RC} \rightarrow \ln |v| = -\frac{t}{RC} + \ln A \rightarrow v = \pm A e^{-t/RC} \rightarrow v(t) = V_0 e^{-t/RC}$$

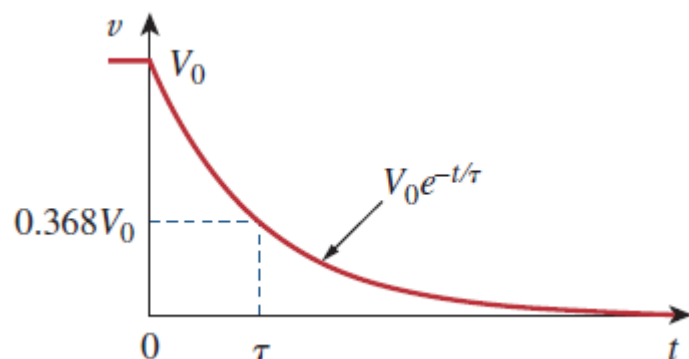


**Figure 7.1**  
A source-free  $RC$  circuit.

# Voltage Response of Source-Free *RC* Circuit

- The voltage response of the source-free *RC* circuit is an exponential decay (指数衰减) of the initial voltage (初始电压)
- Since this is the response of the circuit without any external sources, the response is called the natural response (自由响应)
- The speed of the voltage decay can be related to how long it takes the voltage to drop to  $1/e$  or 36.8% of the initial voltage → This is called the time constant (时间常数) and is represented by  $\tau$

$$V_0 e^{-\tau/RC} = V_0 e^{-1} = 0.368V_0 \Rightarrow \boxed{\tau = RC}$$



**TABLE 7.1**

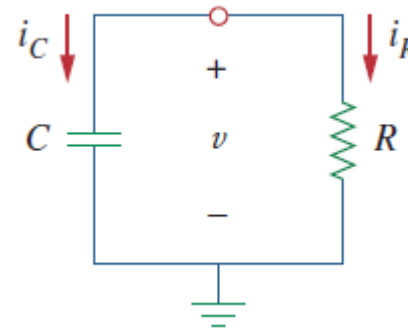
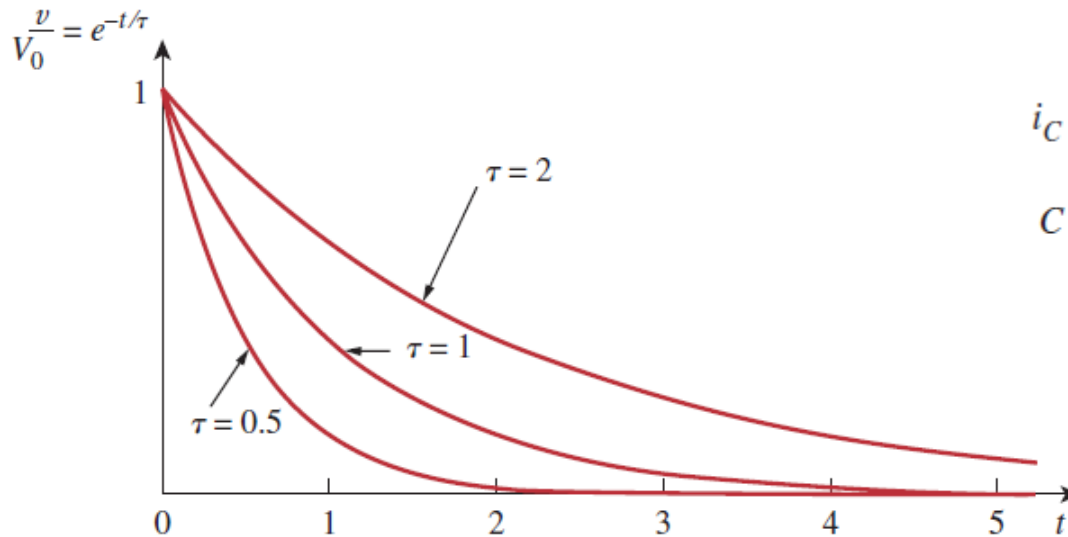
Values of  $v(t)/V_0 = e^{-t/\tau}$ .

| $t$                       | $v(t)/V_0$     |
|---------------------------|----------------|
| $\tau$                    | 0.36788        |
| $2\tau$                   | 0.13534        |
| $3\tau$                   | 0.04979        |
| $4\tau$                   | 0.01832        |
| <b><math>5\tau</math></b> | <b>0.00674</b> |

- After 5-time constants the voltage on the capacitor is less than one percent of its initial value and it is considered to be fully discharged

# Decay Rate of the Source-Free *RC* Circuit

- The smaller the time constant, the more rapidly the voltage decreases, that is, the faster the response
  - A circuit with a small time constant gives a fast response in that it reaches the steady state (or final state) quickly due to quick dissipation of energy stored
  - A circuit with a large time constant gives a slow response because it takes longer to reach steady state



$$\tau = RC$$

$$v(t) = V_0 e^{-t/\tau}$$

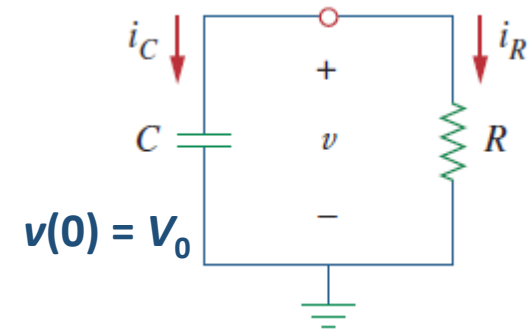
# Current & Energy Response of Source-Free RC Circuit

- The current can be found

$$i_R(t) = \frac{v(t)}{R} = \frac{V_0}{R} e^{-t/\tau}$$

- The power dissipated in the resistor

$$p(t) = vi_R = \frac{V_0^2}{R} e^{-2t/\tau}$$



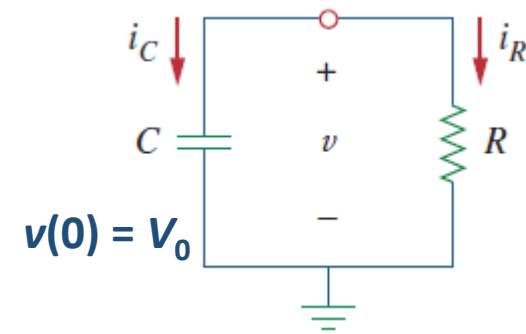
- The energy dissipated by the resistor up to time  $t$  is

$$w_R(t) = \int_0^t p(\lambda) d\lambda = \frac{V_0^2}{R} \int_0^t e^{-2\lambda/\tau} d\lambda = -\frac{1}{2} \tau \cdot \frac{V_0^2}{R} e^{-2\lambda/\tau} \Big|_0^t = \frac{1}{2} C V_0^2 (1 - e^{-2t/\tau})$$

- Notice that as  $t$  approaches infinity, the energy dissipated by the resistor,  $w_R(\infty) \rightarrow 0.5 C V_0^2$ , approaches the initial stored energy in the capacitor
- Therefore, in steady states, the energy that was initially stored in the capacitor is eventually dissipated in the resistor (电容最初存储的能量被电阻消耗)

# Procedures to Analyzing a Source-Free $RC$ Circuit

- As the equation  $v_c(t) = V_0 e^{(-t/\tau)}$  shows, to determine the response of a source-free  $RC$  circuit, we only need to know
  - 1) The **initial voltage** on capacitor  $v(0) = V_0$
  - 2) The **time constant**  $\tau = RC$
- The initial voltage can be obtained from the **steady analysis** of the circuit (treat capacitor as open circuit)
- In finding the time constant  $\tau$ , one may need to use the **Thevenin theorem** to find out the equivalent resistance seen by the capacitor (the circuit seen by the capacitor may contain more than one resistors and dependent sources).
- From the derived voltage  $v_c$ , other variables such as the capacitor current, resistor voltage, current, and dissipated power can be determined



# Example1: The Source-Free $RC$ Circuit

## Example 7.1

In Fig. 7.5, let  $v_C(0) = 15$  V. Find  $v_C$ ,  $v_x$ , and  $i_x$  for  $t > 0$ .

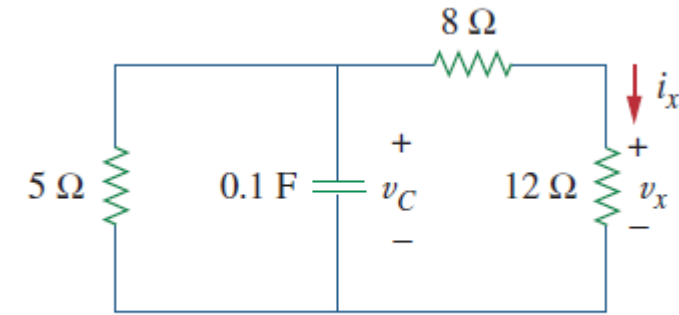


Figure 7.5

## Solution:

- The Thevenin equivalent resistance at the terminals of the capacitor is

$$R_{eq} = 5 \parallel (8 + 12) = 4 \, \Omega$$

- Therefore, the time constant of the  $RC$  circuit is  $\tau = R_{eq}C = 4 \cdot 0.1 = 0.4$  s
- The capacitor voltage is given by  $v_C(t) = V_0 e^{-t/\tau} = 15e^{-2.5t}$
- With this, the voltage  $v_x$  and current  $i_x$  can be found out to be

$$v_x(t) = v_C(t) \cdot \frac{12}{12 + 8} = 9e^{-2.5t} \, \text{V} \quad i_x(t) = \frac{v_x(t)}{12} = 0.75e^{-2.5t} \, \text{A}$$

# Example2: The Source-Free $RC$ Circuit

- The switch in the circuit in Figure was on for a long time, and was switched off  $t = 0$ . Find  $v(t)$  for  $t \geq 0$ .

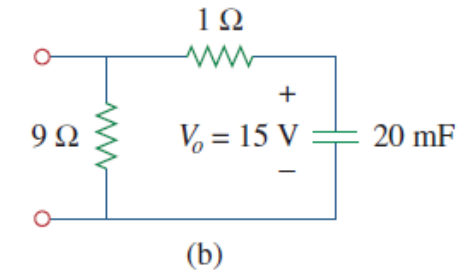
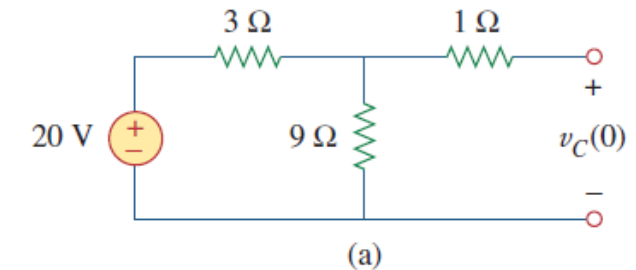
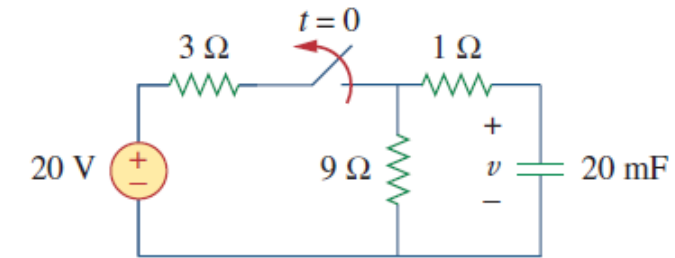
## Solution:

- We can easily find out  $v(0) = 20 \times 9 / (3 + 9) = 15 \text{ V}$
- After the switch is opened, the circuit at the right side of the switch becomes a source-free  $RC$  circuit
- The Thevenin equivalent resistance seen by the capacitor is  $R_{eq} = 1 + 9 = 10 \Omega$  and the time constant of the  $RC$  circuit is

$$\tau = R_{eq}C = 10 \times 20 \times 10^{-3} = 0.2 \text{ s}$$

- So, the capacitor voltage is

$$v(t) = 15e^{-5t} \text{ V}$$



**Figure 7.9**

For Example 7.2: (a)  $t < 0$ , (b)  $t > 0$ .

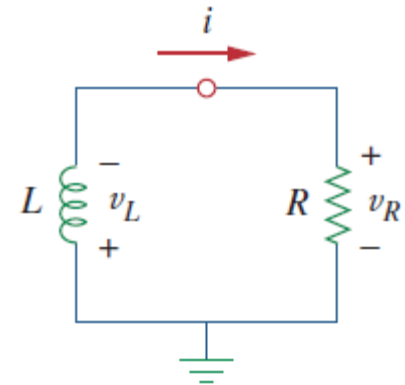
# The Source-Free $RL$ Circuit

- A source-free  $RL$  circuit occurs when the current-carrying inductor is switched to a resistive load. The energy stored in the inductor would start to dissipate as heat on the resistor, representing the discharging process of the inductor
- Note that the resistor and inductor may represent an equivalent resistor and equivalent inductor of complex resistive and inductive networks
- Assume that the initial current of the inductor is  $i(0)=I_0$ . Applying KVL around the loop, we have

$$v_L + v_R = 0 \rightarrow L \frac{di}{dt} + Ri = 0 \rightarrow \frac{di}{dt} + \frac{R}{L}i = 0$$

- Again, the above equation is of first-order and can be solved by the separable equation method, we have

$$\frac{di}{i} = -\frac{R}{L}dt \rightarrow \ln|i| = -\frac{Rt}{L} + \ln A \rightarrow i = \pm Ae^{-Rt/L} \rightarrow i(t) = I_0 e^{-Rt/L}$$



# The Response of the Source-Free *RL* Circuit

- The natural response of *RL* circuit is an exponentially decaying function shown in the inset with a time constant defined as the ratio between the inductance and the resistance, written as

$$\tau = \frac{L}{R} \quad i(t) = I_0 e^{-t/\tau}$$

- The voltage on the resistor

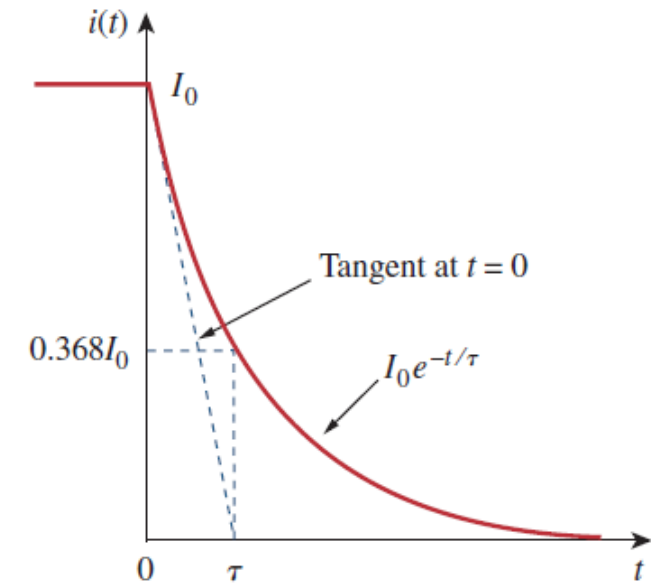
$$v_R(t) = i(t)R = I_0 R e^{-t/\tau}$$

- The power and energy dissipated on the resistor

$$p(t) = v_R i = I_0^2 R e^{-2t/\tau}$$

$$w_R(t) = \int_0^t p(\lambda) d\lambda = I_0^2 R \int_0^t e^{-2\lambda/\tau} d\lambda = -\frac{1}{2} \tau \cdot I_0^2 R e^{-2\lambda/\tau} \Big|_0^t = \frac{1}{2} L I_0^2 (1 - e^{-2t/\tau})$$

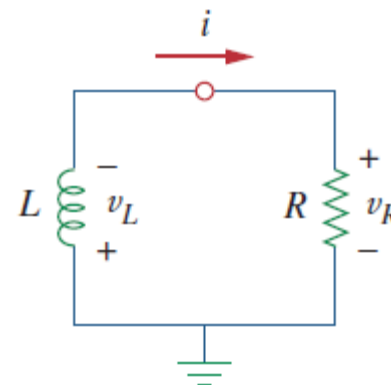
- Similar to the case of *RC*, as  $t \rightarrow \infty$ ,  $w_R(\infty) \rightarrow 0.5 L I_0^2$ , the dissipated energy approaching that stored in the inductor (电感最初存储的能量被电阻完全消耗)



# Procedures to Analyzing a Source-Free *RL* Circuit

- Similar to the case of *RC* circuit, the key to solving a source-free *RL* circuit is to find
  - 1) The **initial current** through the inductor  $i(0) = I_0$
  - 2) The **time constant**  $\tau = L/R$
- With these numbers, the inductor current for  $t > 0$  is given by

$$i_L(t) = i(t) = I_0 e^{-t/\tau}$$



- From the inductor current, other variables such the voltage, current, energy of the resistor can be determined
- The resistance  $R$  might be the **Thevenin equivalent resistance** at the terminals of the inductor
- When there are dependent sources in the circuit, the inductor should be replaced with an independent current source to facilitate the calculation of the resistance

# Example1: The Source-Free *RL* Circuit

- Assuming that  $i(0) = 10$  A, calculate  $i(t)$  and  $i_x(t)$  in the circuit below

## Solution1:

- Write KCL for the node at the top

$$i + \frac{0.5}{2} \frac{di(t)}{dt} + \frac{0.5di(t)/dt - 3i}{4} = 0 \Rightarrow \frac{di(t)}{dt} = -\frac{2}{3}i(t)$$

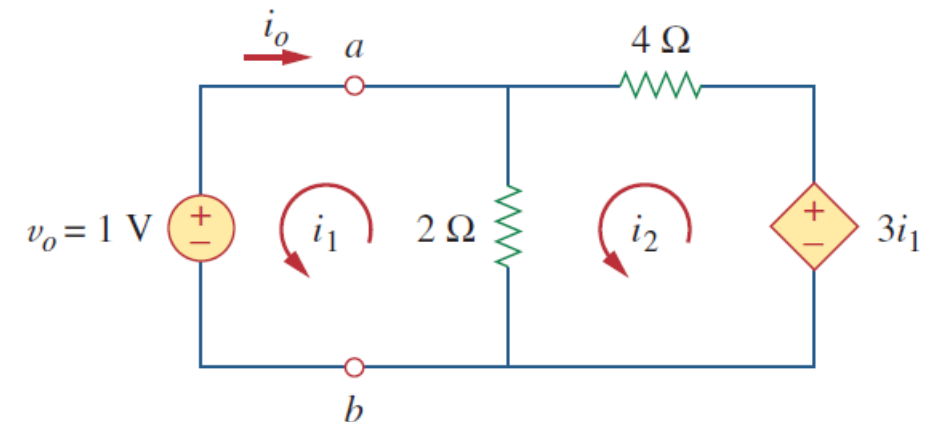
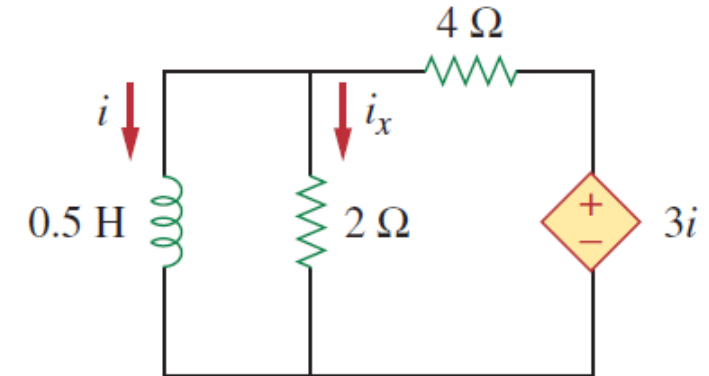
$$i(t) = i(0)e^{-(2/3)t} = 10e^{-(2/3)t} \text{ A}, \quad t > 0$$

$$v = L \frac{di}{dt} = 0.5(10) \left( -\frac{2}{3} \right) e^{-(2/3)t} = -\frac{10}{3} e^{-(2/3)t} \text{ V} \quad i_x(t) = \frac{v}{2} = -1.6667 e^{-(2/3)t} \text{ A}, \quad t > 0$$

## Solution2:

- Calculate the  $R_{th} = 1/3$
- The time constant is  $\tau = \frac{L}{R_{eq}} = \frac{1/2}{1/3} = \frac{3}{2} \text{ s}$
- The current through the inductor is

$$i(t) = i(0)e^{-t/\tau} = 10e^{-(2/3)t} \text{ A}, \quad t > 0$$

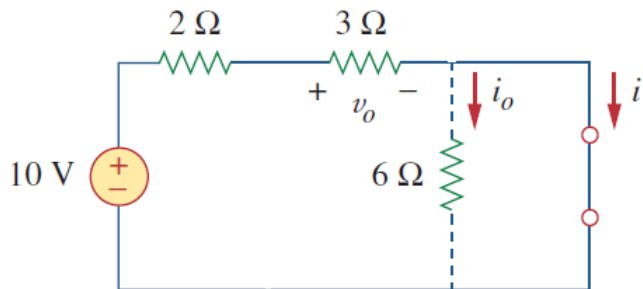


# Example2: The Source-Free *RL* Circuit

- In the circuit shown below, find  $i_o(t)$ ,  $v_o(t)$  and  $i(t)$  for  $t > 0$ , assuming that the switch was open for a long time

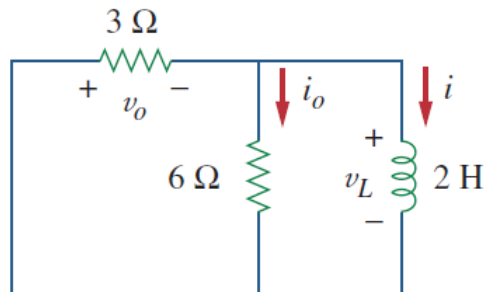
## Solution:

- Initial current through the 2H inductor



$$i(0) = 2\text{A}$$

- The time constant



$$\tau = L / R$$

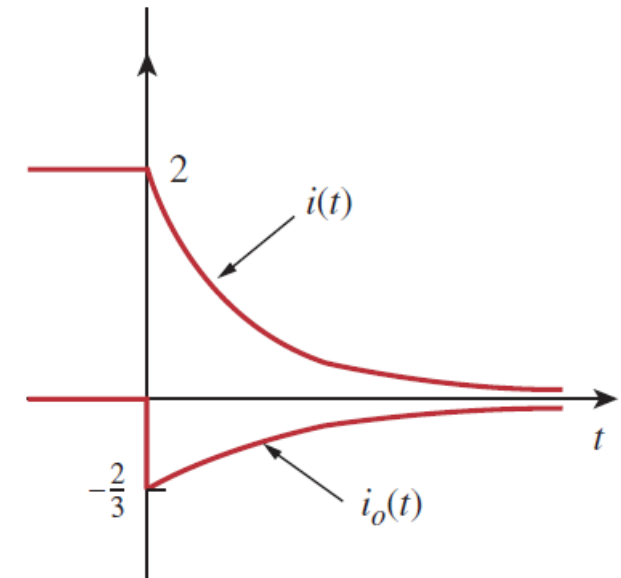
$$= 2\text{ H} / (3\text{ }\Omega \parallel 6\text{ }\Omega)$$

$$= 1\text{ s}$$

$$\tau = \frac{L}{R_{eq}} = \frac{2}{2} = 1\text{ s}, i(t) = \begin{cases} 2\text{ A}, & t < 0 \\ 2e^{-t}\text{ A}, & t \geq 0 \end{cases}$$

$$v_o(t) = \begin{cases} 10 \cdot \frac{3}{2+3}\text{ V}, & t < 0 \\ -L \frac{di}{dt}, & t \geq 0 \end{cases} = \begin{cases} 6\text{ V}, & t < 0 \\ 4e^{-t}\text{ V}, & t \geq 0 \end{cases}$$

$$i_o(t) = \frac{v_L}{6} = L \frac{di}{dt} \frac{1}{6} = \begin{cases} 0, & t < 0 \\ -\frac{2}{3}e^{-t}\text{ A}, & t \geq 0 \end{cases}$$

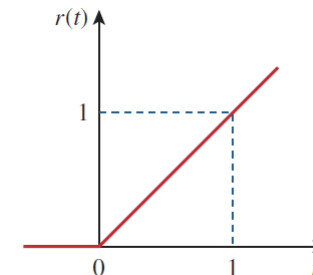
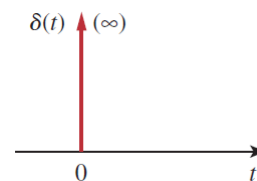
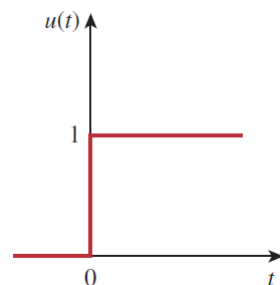


# Outline

- Introduction
- The Source-Free *RC* Circuit
- The Source-Free *RL* Circuit
- **Singularity Functions**
- **Step Response of an *RC* Circuit**
- **Step Response of an *RL* Circuit**

# Singularity Functions

- Before we consider the response of a circuit to an external voltage or current, we need to cover some mathematical concepts that will aid our understanding of transient analysis (瞬态分析)
- Singularity functions (奇异函数) (also called switching functions) are very useful in circuit analysis
  - They serve as good approximations to the switching signals (开关信号) that arise in circuits with switching operations (e.g., on or off).
  - They are helpful in the compact description of some circuit phenomena, especially the step response of  $RC$  or  $RL$  circuits to be discussed soon
- The three most commonly used singularity functions are the unit step (单位阶跃), the unit impulse (单位冲激), and the unit ramp (单位斜坡) functions

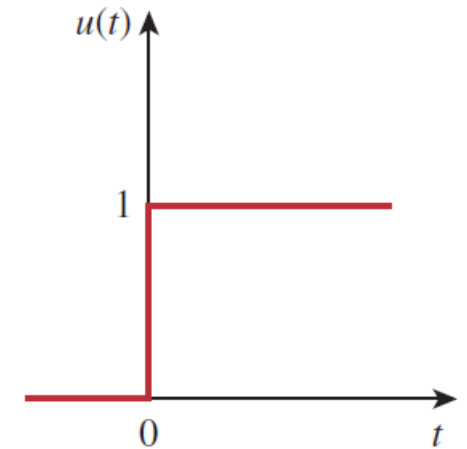


# Singularity Functions: Unit Step (I)

- A **step function** (阶跃函数) is one that maintains a constant value before a certain time and then changes to another constant afterwards
- We use the step function to represent an abrupt change in voltage or current
- The prototypical form is the unit step function  $u(t)$  which is 0 for negative values of  $t$  and 1 for positive values of  $t$ . The unit step function can be expressed as

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

- The unit step function is undefined at  $t = 0$  where it changes abruptly from 0 to 1
- Other step functions can be represented as linear combinations of translations of the unit step function  $u(t)$



**Figure 7.23**  
The unit step function.

# Singularity Functions: Unit Step (II)

- The switching time may be shifted to  $t = t_0$  by

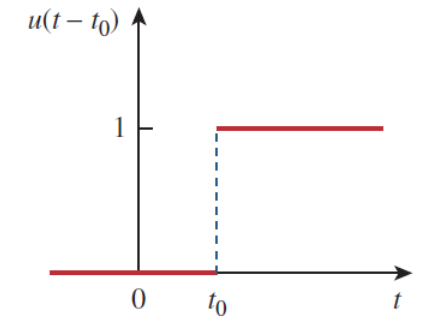
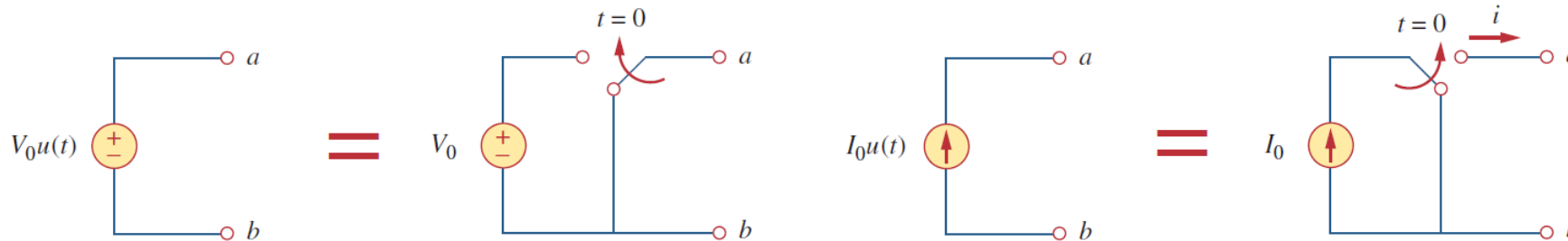
$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$$

➤ which is the same as saying that  $u(t)$  is delayed by  $t_0$  seconds

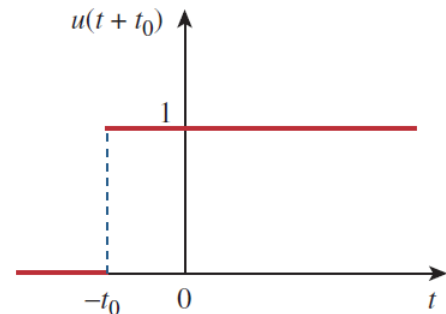
- Similarly, the switching time may be shifted to  $t = -t_0$  by

$$u(t + t_0) = \begin{cases} 0, & t < -t_0 \\ 1, & t > -t_0 \end{cases}$$

- The unit step function has an equivalent circuit to represent when it is used to switch on a source



(a)



(b)

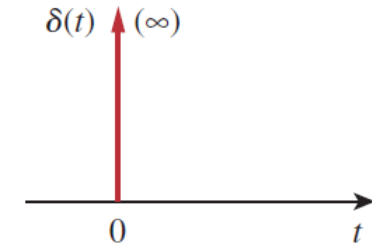
**Figure 7.24**

(a) The unit step function delayed by  $t_0$ .  
 (b) the unit step advanced by  $t_0$ .

# Singularity Functions: Unit Pulse

- The derivative of  $u(t)$  is the unit impulse function (单位冲激函数), also known as the delta function (狄拉克函数)

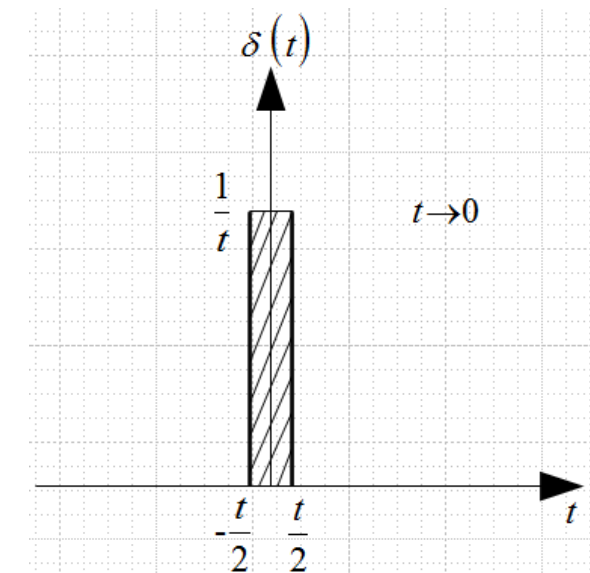
$$\delta(t) = \frac{d}{dt}u(t) = \begin{cases} 0, & t < 0 \\ \text{Undefined}, & t = 0 \\ 0, & t > 0 \end{cases}$$



**Figure 7.27**  
The unit impulse function.

- Impulsive currents and voltages occur in electric circuits as a result of switching operations or impulsive sources. The unit impulse function is not physically realizable, but it is a useful mathematical tool
- It can be visualized as a very short duration pulse of unit area and mathematically expressed as

$$\int_{0^-}^{0^+} \delta(t) dt = 1$$



# Singularity Functions: Unit Ramp

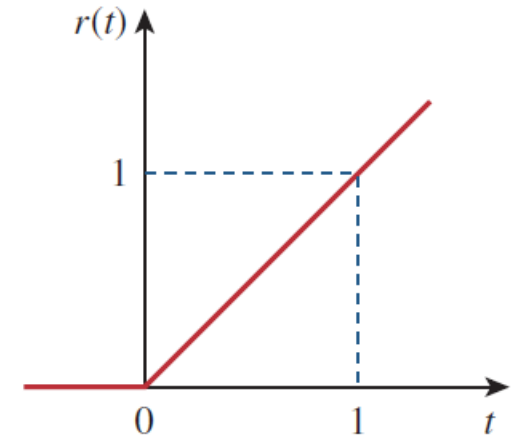
- Integration of  $u(t)$  results in the unit ramp function (单位斜坡函数)

$$r(t) = \int_{-\infty}^t u(\lambda) d\lambda = tu(t) \quad \text{or} \quad r(t) = \begin{cases} 0, & t \leq 0 \\ t, & t \geq 0 \end{cases}$$

- The unit ramp function is zero for negative values of  $t$  and has a unit slope for positive values of  $t$ . In general, a ramp is a function that changes at a constant rate
- We should keep in mind that the three singularity functions (impulse, step, and ramp) are related as

$$\delta(t) = \frac{du(t)}{dt}, \quad u(t) = \frac{dr(t)}{dt}$$

$$u(t) = \int_{-\infty}^t \delta(\lambda) d\lambda, \quad r(t) = \int_{-\infty}^t u(\lambda) d\lambda$$

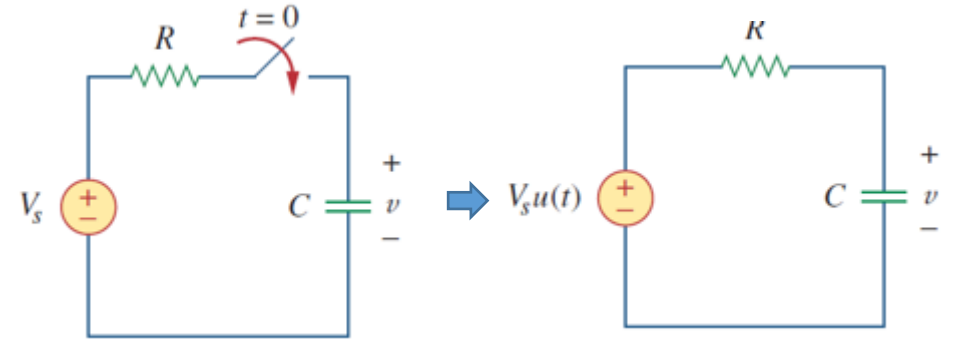


**Figure 7.29**  
The unit ramp function.

# Step Response of an $RC$ Circuit (I)

- When the dc source is suddenly applied, the voltage or current source can be modeled as a step function, and the response is called a **step response** (阶跃响应)
- Assume that the initial voltage across capacitor is  $v(0) = V_0$ . Applying KCL, yielding

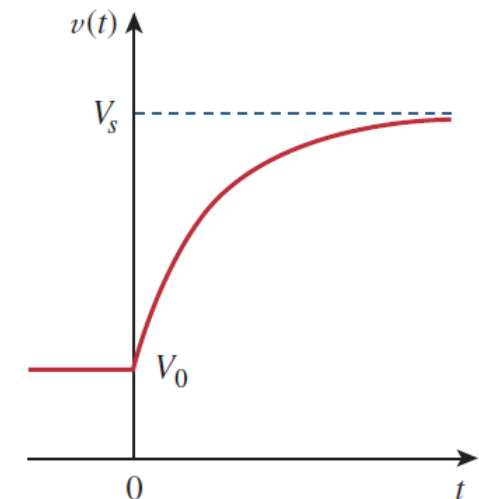
$$C \frac{dv}{dt} + \frac{v - V_s u(t)}{R} = 0 \Rightarrow \frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC}$$



- With **general solution** method, we can solve for the solution (with initial condition) and get

$$\boxed{\frac{dy}{dx} + P(x)y = Q(x)} \quad \boxed{I(x) = e^{\int P(x)dx}} \quad \boxed{y = \frac{1}{I(x)} \left[ \int I(x)Q(x)dx + C \right]}$$

$$\boxed{v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau}, & t > 0 \end{cases}} \quad \tau = RC$$

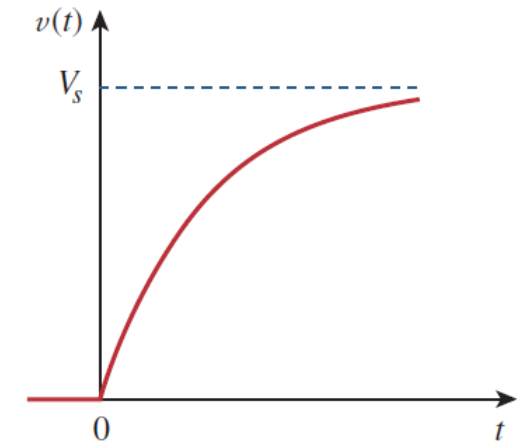


# Step Response of an *RC* Circuit (II)

- This is known as the **complete response** (or total response, 全响应) of the *RC* circuit to a sudden application of a dc voltage source
- If the capacitor is uncharged initially, i.e.,  $V_0 = 0$ , we have

$$v(t) = \begin{cases} 0, & t < 0 \\ V_s(1 - e^{-t/\tau}), & t > 0 \end{cases}$$

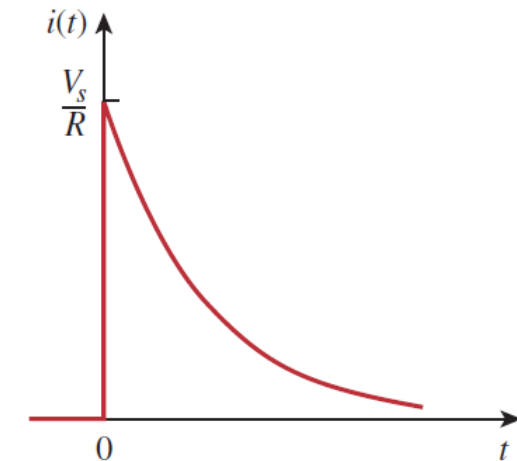
$$\Rightarrow v(t) = V_s(1 - e^{-t/\tau})u(t)$$



- This is the step response of the *RC* circuit when the capacitor is initially uncharged. From this, the capacitor current can be found as

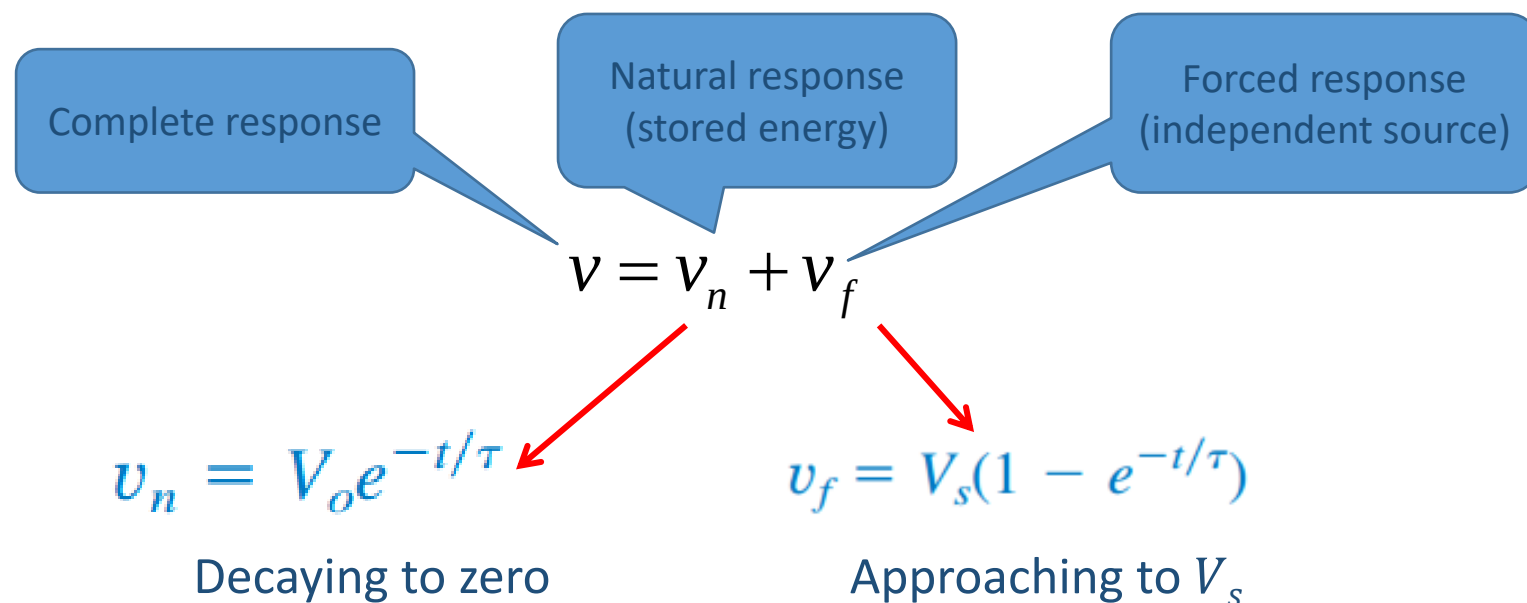
$$i(t) = C \frac{dv}{dt} = \frac{C}{\tau} V_s e^{-t/\tau}, \quad \tau = RC, \quad t > 0$$

$$\Rightarrow i(t) = \frac{V_s}{R} e^{-t/\tau} u(t)$$



# Understanding the Response of an *RC* Circuit (I)

- Obtaining differential equations and solving for them can of course lead us to the circuit response solutions. However, there is a systematic approach for finding the step response of an *RC* or *RL* circuit which is to break the complete response into two separate responses
- Two ways of decomposition exist: 1) “a natural response (自由响应) and a forced response (受迫响应)”, and 2) “a transient response (瞬态响应) and a steady-state response (稳态响应).”
- Perspective 1):**



# Understanding the Response of an *RC* Circuit (II)

- Two ways of decomposition exist: 1) “a natural response (自由响应) and a forced response (受迫响应)”, and 2) “a transient response (瞬态响应) and a steady-state response (稳态响应).”
- Perspective 2):**

Complete response = transient response + steady-state response  

temporary part
permanent part

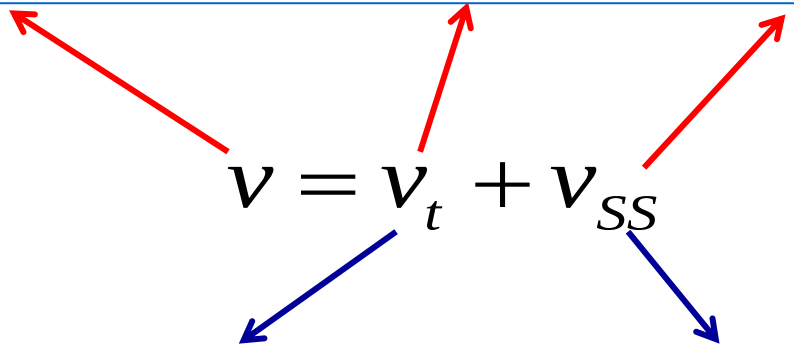
  

$$v = v_t + v_{ss}$$

(Decaying to 0)

 $v_t(t) = (V_0 - V_S)e^{-t/\tau}$

$v_{ss} = v(\infty) = V_S$  (constant)



- The transient response  $v_t(t)$  represent the part in the complete response that will die out eventually
- The steady-state response  $V_{ss}$  represents the response of the circuit to external excitations that was applied sufficiently long time ago

# General Approach to Solving the *RC* Circuit

- From these perspectives, the step response of an *RC* circuit can be written as

$$v(t) = v_n + v_f = v(0)e^{-t/\tau} + v(\infty)(1 - e^{-t/\tau})$$

or

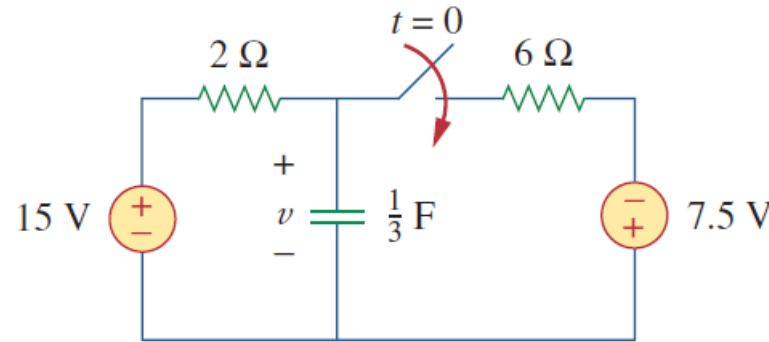
$$v(t) = v_{ss} + v_t = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

- Where  $v(0)$  is the initial voltage and  $v(\infty)$  is the final or steady-state value. Thus, to find the step response of an *RC* circuit requires the initial capacitor voltage  $v(0)$ , the final capacitor voltage  $v(\infty)$  and the time constant  $\tau$
- Thus, to find the step response of an *RC* circuit requires
  - 1) the initial capacitor voltage  $v(0)$
  - 2) the final capacitor voltage  $v(\infty)$
  - 3) the time constant  $\tau$
- When excitation is applied at  $t = t_0$  instead of  $t = 0$ , the response will be accordingly delayed by  $t_0$

$$v(t) = v_n + v_f = v(t_0)e^{-(t-t_0)/\tau} + v(\infty)(1 - e^{-(t-t_0)/\tau})$$

# Example1: Step Response of an *RC* Circuit

- Find  $v(t)$  for  $t > 0$  in the circuit of below. Assume the switch has been open for a long time and is closed at  $t = 0$ . Calculate  $v(t)$  at  $t = 0.5$  s



## Solution:

- The initial capacitor voltage  $v(0) = V_0 = 15$  V.
- The final capacitor voltage  $v(\infty) = 15 \cdot 6 / (2+6) - 7.5 \cdot 2 / (2+6) = 9.375$  V
- The time constant  $\tau = RC = (2 \Omega \parallel 6 \Omega) \cdot 1/3 \text{ F} = 0.5$  s

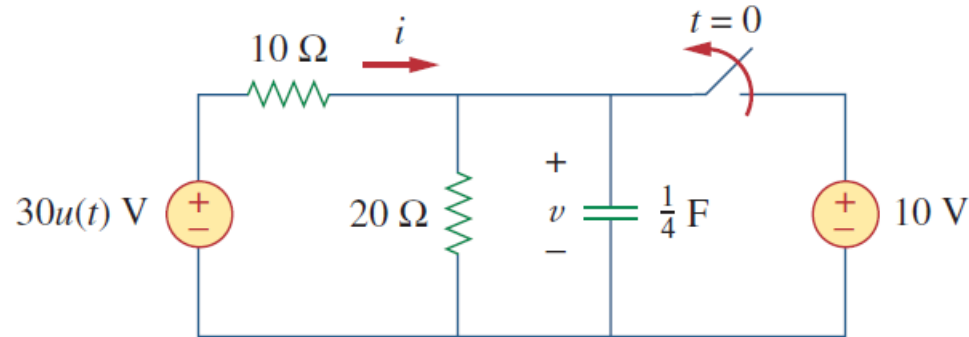
The capacitor voltage

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau} = (9.375 + 5.625e^{-t/0.5}) \text{ V}$$

$$v(0.5) = 9.375 + 5.625e^{-0.5/0.5} = \mathbf{11.44 \text{ V}}$$

## Example2: Step Response of an RC Circuit

- In the following figure, the switch has been closed for a long time and is opened at  $t = 0$ . Find  $i(t)$  and  $v(t)$  for  $t > 0$ .



### Solution:

- The initial capacitor voltage is  $v(0) = V_0 = 10 \text{ V}$
- The steady-state capacitor voltage is  $v(\infty) = 30 \cdot 20 / (10 + 20) = 20 \text{ V}$
- The time constant is  $\tau = RC = (10 \Omega \parallel 20 \Omega) \cdot (1/4) = 5/3 \text{ s}$

Therefore, for  $t > 0$

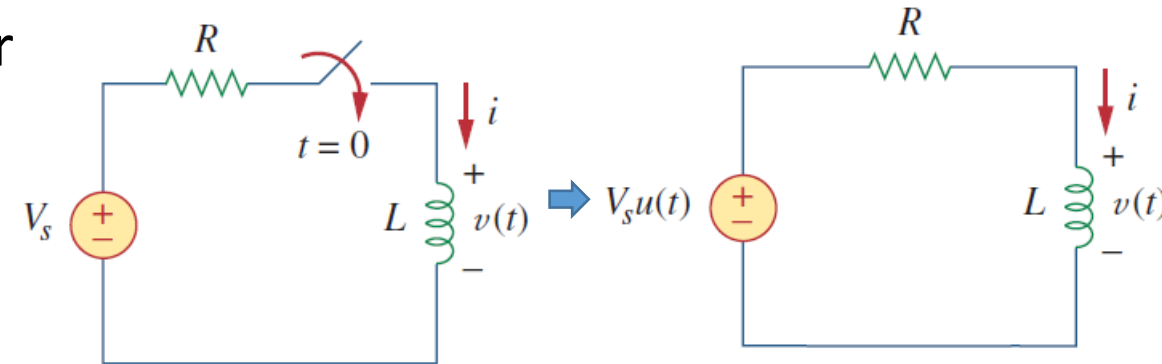
$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau} = (20 - 10e^{-0.6t}) \text{ V}$$

$$i(t) = \frac{v}{20} + C \frac{dv}{dt} = 1 + e^{-0.6t}$$

# Step Response of an $RL$ Circuit (I)

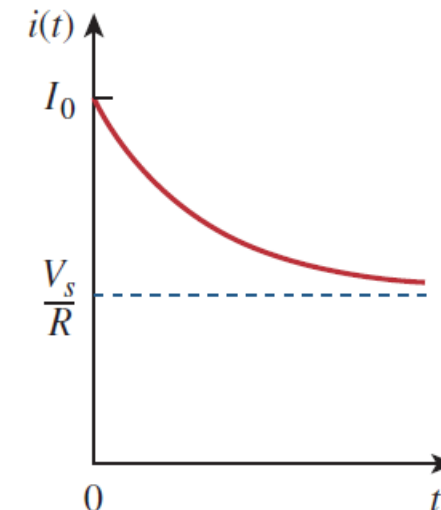
- Now we consider the  $RL$  circuit shown right, where  $V_s$  is a constant dc voltage source. We can find the inductor current  $i$  as a function of time
- Assume that the initial current across inductor is  $i(0) = I_0$ . Applying KVL, yielding

$$L \frac{di}{dt} + iR = V_s u(t) \rightarrow \frac{di}{dt} + i \frac{R}{L} = \frac{V_s}{L} \quad \text{for } t > 0$$



- With separable equation method, we can solve for the solution (with initial condition) and get the complete step response as

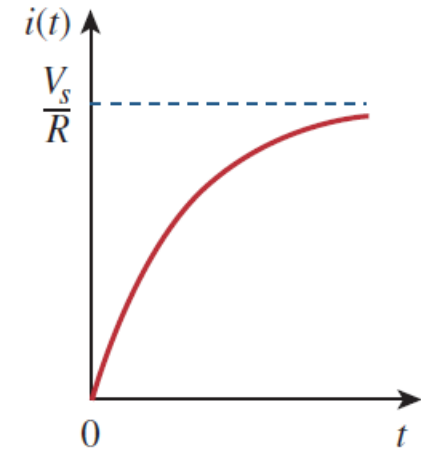
$$i(t) = \begin{cases} I_0, & t < 0 \\ \frac{V_s}{R} + \left( I_0 - \frac{V_s}{R} \right) e^{-t/\tau}, & t > 0 \end{cases} \quad \tau = \frac{L}{R}$$



# Step Response of an *RL* Circuit (II)

- If there is no initial inductor current, i.e.,  $I_0 = 0$ , we have

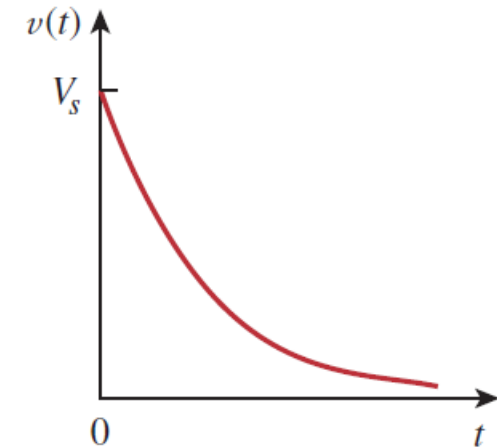
$$i(t) = \begin{cases} 0, & t < 0 \\ \frac{V_s}{R}(1 - e^{-t/\tau}), & t > 0 \end{cases} \Rightarrow \boxed{i(t) = \frac{V_s}{R}(1 - e^{-t/\tau})u(t)}$$



- This is the step response of the *RL* circuit when the inductor has no initial current
- From this, the inductor voltage can be found as

$$v(t) = L \frac{di}{dt} = V_s \frac{L}{\tau R} e^{-t/\tau}, \quad \tau = \frac{L}{R}, \quad t > 0$$

$$\Rightarrow v(t) = V_s e^{-t/\tau} u(t)$$



# Understanding the Step Response of RL Circuit

- We can also apply the systematic approach of decomposition to find the step response of the RL circuit. Let the response be the sum of the **transient response** (瞬态响应) and the **steady-state response** (稳态响应)

$$\dot{i} = \dot{i}_t + \dot{i}_{ss}$$

$$i_t = \left( I_0 - \frac{V_s}{R} \right) e^{-t/\tau} = [i(0) - i(\infty)] e^{-t/\tau}$$
$$i_{ss} = i(\infty) = \frac{V_s}{R}$$

- The full response is determined by three values: 1) *initial value*  $i(0)$ ; 2) *final value*  $i(\infty)$ , and 3) *time constant*  $\tau$

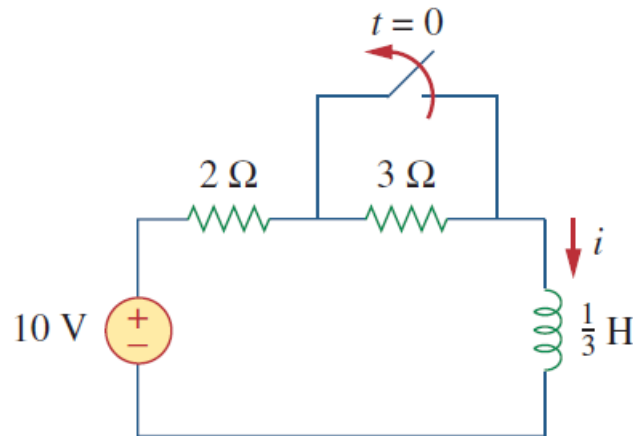
$$i(t) = i_{ss} + i_t = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$

- When the switching occurs at  $t = t_0$ , the step response becomes

$$i(t) = i_{ss} + i_t = i(\infty) + [i(t_0) - i(\infty)] e^{-(t-t_0)/\tau}$$

# Example1: Step Response of an *RL* Circuit

- Find  $i(t)$  in the circuit of the following figure for  $t > 0$ . Assume that the switch has been closed for a long time



## Solution:

The switch is opened at time  $t = 0$ , which creates a step function for the circuit. The task is to find the step response of an *RL* circuit, with  $R = 2+3 = 5 \Omega$ ,  $L = 1/3 \text{ H}$ .

- 1) The initial inductor current is  $i(0) = I_0 = 10/2 = 5 \text{ A}$
- 2) The steady-state inductor current is  $i(\infty) = 10/(2+3) = 2 \text{ A}$
- 3) The time constant is  $\tau = L/R = 1/15 \text{ s}$

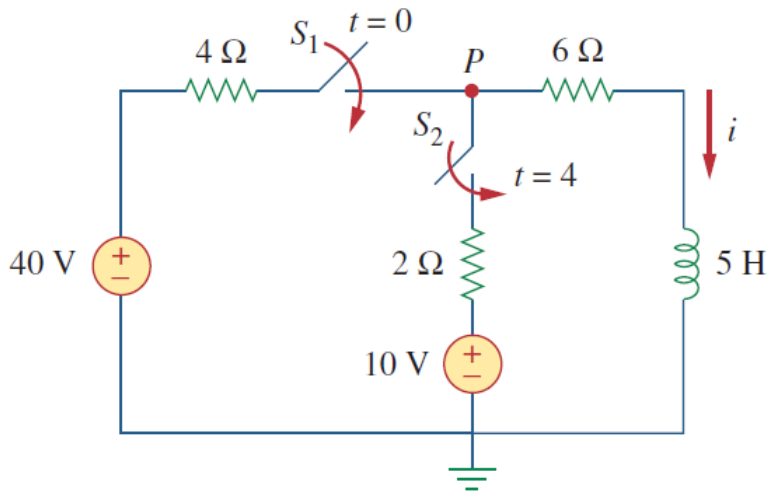
Therefore, for  $t > 0$

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau} = 2 + 3e^{-15t} \text{ A}$$

*\*Note: We can use KVL to check the solution:  $Ldi/dt + iR = V_s$*

# Example2: Step Response of an *RL* Circuit

- At  $t = 0$ , switch 1 in figure is closed, and switch 2 is closed 4 s later. Find  $i(t)$  for  $t > 0$ . Calculate  $i$  for  $t = 2$  s and  $t = 5$  s. Assume  $i(0) = 0$



## Solution:

We need to consider the three time intervals:  $t < 0$ ,  $0 < t < 4$ , and  $t > 4$  separately.

1. For  $t < 0$ , both switches are open, no current flows through the inductor,  $i(t) = 0$

2. At  $t = 0$ , switch  $S_1$  is on. The resulting circuit has a time constant  $\tau = L/R = 5\text{H}/(4\Omega + 6\Omega) = 0.5$  s, with a steady current  $i(\infty) = 40/10 = 4$  A. thus  $i(t_0) = 4 - 4e^{-t/0.5}$  A.

3. At time  $t = 4$  s, switch  $S_2$  is closed. The initial current is  $i(4) = 4 - 4e^{-8} \approx 4$  A ( $4\text{ s} \gg 5\tau$ ). The circuit has now a time constant  $\tau' = L/R' = 5\text{H}/(6+4\parallel 2) = 15/22$  s. The final current is

$i(\infty) = 40 \cdot (2\parallel 6) / [(2\parallel 6) + 4] / 6 + 10 \cdot (4\parallel 6) / [(4\parallel 6) + 2] / 6 = 30/11$  A. Thus

$$i(t) = 2.727 + (4 - 2.727)e^{-(t-4)/\tau}, \quad \tau = \frac{15}{22}$$

$$= 2.727 + 1.273e^{-1.4667(t-4)}, \quad t \geq 4$$

$$i(t) = \begin{cases} 0, & t \leq 0 \\ 4(1 - e^{-2t}), & 0 \leq t \leq 4 \\ 2.727 + 1.273e^{-1.4667(t-4)}, & t \geq 4 \end{cases}$$

At  $t = 2$ ,

$$i(2) = 4(1 - e^{-4}) = 3.93 \text{ A}$$

At  $t = 5$ ,

$$i(5) = 2.727 + 1.273e^{-1.4667} = 3.02 \text{ A}$$

# Summary (I)

- The analysis in this lecture is applicable to any circuit that can be reduced to an equivalent circuit comprising a resistor and a single energy-storage element (inductor or capacitor)
  - Such a circuit is of first-order because its behavior is described by a first-order differential equation
  - In the steady-state dc conditions, the capacitor is an open circuit while the inductor is a shorted circuit
- The natural response occurs when no independent source is present. It has the general form  $x(t) = x(0)e^{-t/\tau}$  where  $x$  represents current through or voltage across a resistor, a capacitor, or an inductor;  $x(0)$  is the initial value of  $x$  at  $t = 0$ 
  - Because most practical resistors, capacitors, and inductors always have losses, the natural response is a transient response, i.e. it dies out with time
  - The time constant  $\tau$  is the time required for a response to decay to  $1/e$  of its initial value. For  $RC$  circuits,  $\tau = RC$  and for  $RL$  circuits,  $\tau = L/R$

# Summary (II)

- The total or complete response consists of the steady-state response and the transient response
  - The steady-state response is the behavior of the circuit after an independent source has been applied for a long time.
  - The transient response is the component of the complete response that dies out with time
- The step response is the response of the circuit to a sudden application of a dc current or voltage. Finding the step response of a first-order circuit requires: 1) the initial value  $x(0^+)$ ; 2) the final value  $x(\infty)$ ; and 3) the time constant  $\tau$ . With these three items, we obtain the step response as

$$x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$$

- Acknowledgement: This Lecture Note is partially based on materials from Prof. Chenchang Zan and partially adapted from the notes of the book “Fundamentals of Electric Circuits” by C. K. Alexander and M. N. O. Sadiku